SYSTEMS ANALYSIS & DESIGN

Systems, Information & Information Systems

Foundational concepts, the systems analysis process, information quality, and the major types of information systems used in organizations.

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CORE CONCEPT

What Is a System?

Etymology. The word "System" is derived from the Greek word Systema, meaning an organized relationship between any set of components to achieve some common cause or objective.

Definition. A system is an orderly grouping of interdependent components linked together according to a plan to achieve a specific goal. Every part of a system depends on and affects the others, and together they work toward one shared purpose.

Three Basic Constraints of a System

1. Must have structure and behavior designed to achieve a predefined objective.
2. Interconnectivity and interdependence must exist among the system's components.
3. The objectives of the organization outrank the objectives of its individual subsystems.

BUILDING BLOCKS OF A SYSTEM

Key Components of a System (I)

1
Input

The data, materials, and resources that enter the system from its environment, forming the raw starting point for every process.

2
Process

The set of operations and activities that transform raw inputs into meaningful, usable outputs.

3
Output

The final product, service, or result the system delivers after processing — the tangible outcome of its purpose.

4
Feedback

Information about the system's output that is looped back in to monitor, evaluate, and improve future performance.

Key Components of a System (II)

5 Control Mechanism

The rules, checks, and management processes that regulate system performance and keep it aligned with its objectives.

6 Boundary

The defined limit that separates the system from its external environment and marks what is inside versus outside its scope.

7 Environment

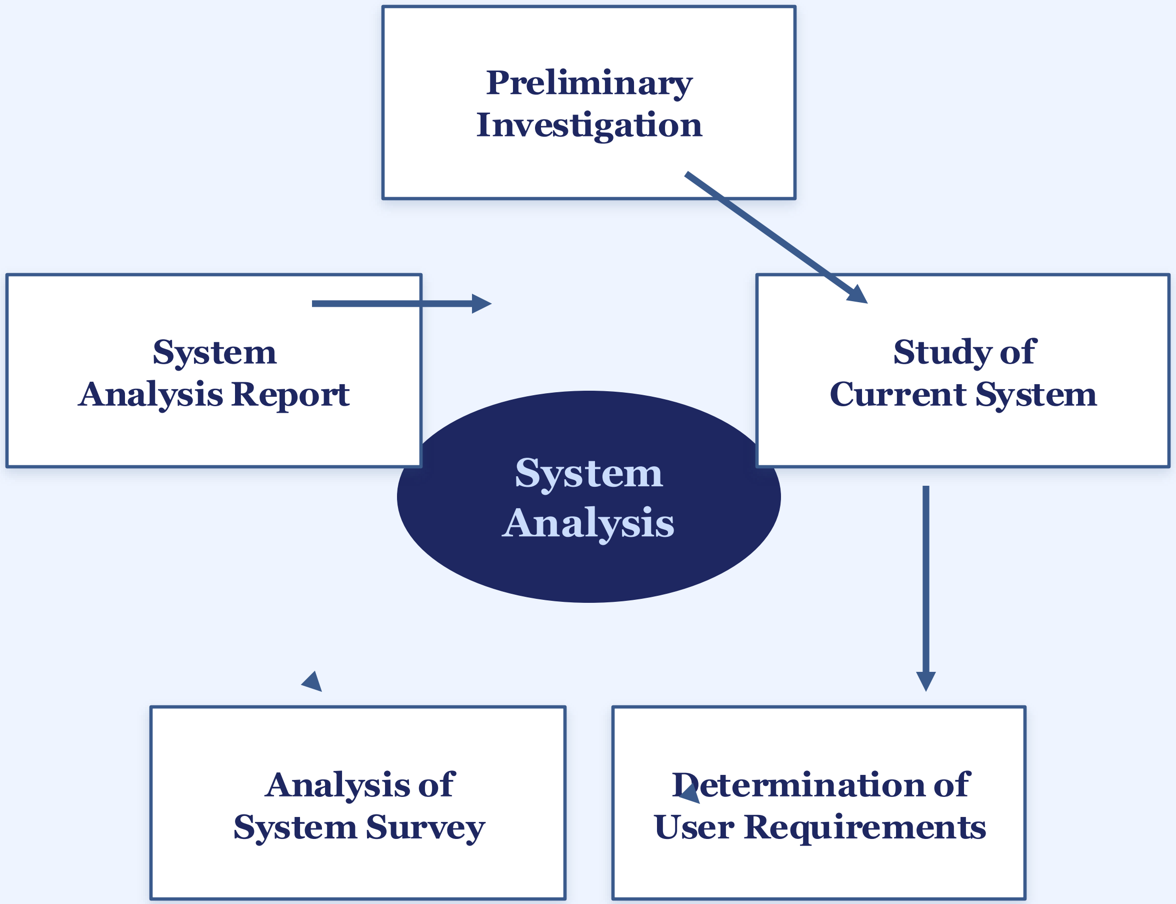
The external factors — social, economic, technological, or regulatory — that surround and influence the system's behavior.

What Is Systems Analysis?

Systems analysis describes in detail the "what" that a system must do to satisfy a need or solve a problem. It is the very first step in any system development effort and the critical phase where developers come together to understand the problem, the needs, and the objectives of the project before any design work begins.

In short, analysis defines the problem clearly enough that design can begin building the right solution — getting this stage wrong means every later stage inherits the mistake.

The Systems Analysis Cycle



The Five Stages

Preliminary Investigation — first look at the request and problem.

Study of Current System — examine how things work today.

Determine User Requirements — capture what users need.

Analysis of System Survey — evaluate findings from the study.

System Analysis Report — document conclusions, then repeat.

IN PRACTICE

Key Aspects of Systems Analysis (I)

1 Problem Identification

Identifying the issues the system must address — automating a process, improving data management, or the user experience. The critical first step.

2 Requirements Gathering

Once the problem is identified, the next step is to gather and write down requirements — communicating with the customer and developer about how the system is to be designed.

3 Feasibility Study

Before development begins, it is important to check the feasibility of the project, evaluating its technical, operational, and financial aspects to confirm the proposed solution is achievable.

IN PRACTICE

Key Aspects of Systems Analysis (II)

4 Analysis & Modeling

To get a deep insight into the system, analysts develop various models such as Data Flow Diagrams (DFD), Use Cases, and Entity-Relationship (ER) diagrams. These models help the customer visualize the system and its interactions.

5 Scope Definition

Defining the scope of the system is important to prevent adding excessive features and to keep the project within its limits — identifying what is part of the system and what is not.

Types of Systems (I)

Open System

An open system continuously interacts with its external environment by exchanging information, energy, or materials. It is adaptable and influenced by external factors such as changes in regulations, technology, or market conditions, and evolves over time based on feedback.

e.g. *Universities interacting with students and policy; businesses adjusting to customers and competitors.*

Closed System

A closed system does not interact with its external environment and operates independently. It does not exchange matter, energy, or information with its surroundings. True closed systems are rare in reality, since most systems interact with their environment to some degree.

e.g. *A sealed chemical reaction; an isolated laboratory experiment protected from contamination.*

Types of Systems (II)

Deterministic System

A deterministic system follows a predefined set of rules or processes, leading to predictable outcomes. Given the same input, it will always produce the same result, with no uncertainty or randomness — making it reliable for decision-making and automation.

e.g. *A vending machine dispensing a product; a fixed-sequence traffic light controlling vehicle flow.*

Probabilistic System

A probabilistic system operates under conditions of uncertainty, meaning its outcomes are not always the same for a given input. It depends on probability distributions, external conditions, or random variables, and is often analyzed with statistical models.

e.g. *Stock market fluctuations; weather forecasting based on historical and atmospheric data.*

CLASSIFICATION

Types of Systems (III)

Physical System

A physical system consists of tangible, material components that can be seen, touched, and measured. It exists in the real world and operates based on physical laws, and may be natural or man-made — often used in engineering and manufacturing.

e.g. *A computer's hardware components such as processor and memory; a car engine's mechanical parts.*

Abstract System

An abstract system consists of theoretical models, concepts, or software-based structures that help analyze, design, or simulate real-world scenarios. It has no physical form but plays a crucial role in decision-making, computation, and problem-solving.

e.g. *A mathematical model predicting the behavior of real-world systems using equations and simulations.*

CLASSIFICATION

Types of Systems (IV)

Complex System

A complex system consists of multiple interconnected components that interact in unpredictable or intricate ways. These systems often exhibit emergent behaviors, meaning the whole system's properties cannot be fully understood just by analyzing its individual components in isolation. Studying a complex system therefore requires looking at relationships and interactions, not just the parts on their own.

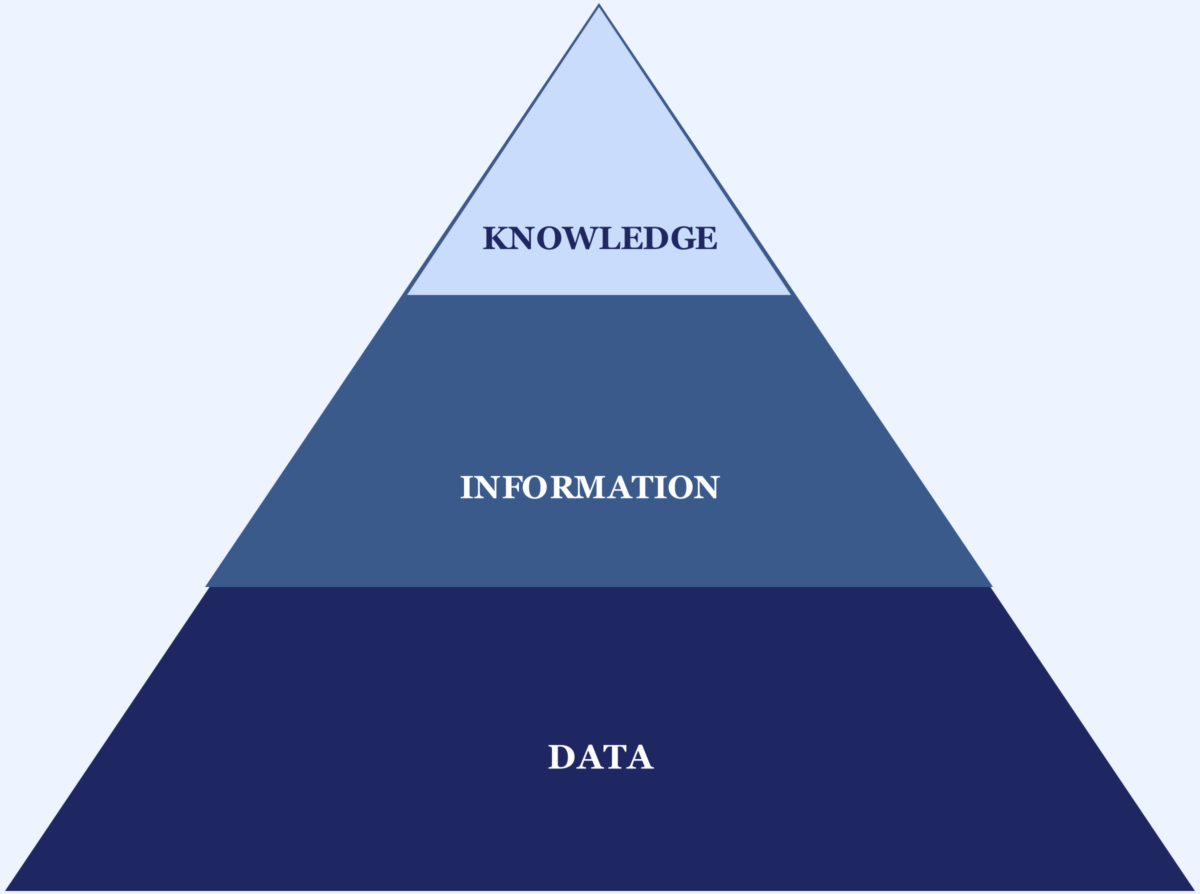
e.g. *A modern transportation system, where roads, traffic lights, and various modes of transport interact dynamically.*

All Seven Types, at a Glance

- Open — exchanges with environment
- Closed — self-contained, independent
- Deterministic — predictable outcomes
- Probabilistic — outcomes uncertain
- Physical — tangible components
- Abstract — theoretical / conceptual
- Complex — emergent, interconnected

THE HIERARCHY

Data, Information & Knowledge



Data

Raw, unprocessed facts and figures that have no context or purposeful meaning on their own. Data is generally used by machines and is useless unless it is processed to create information.

Information

Processed data that is useful, so that the decision-maker may take necessary action based on it.

Knowledge

Data and information that is refined further based on the facts, truths, beliefs, judgments, experiences, and expertise of the recipient.

Quality of Information (I)

1 Accuracy

Information should be free from errors and correctly represent real-world facts. Inaccurate information can lead to poor decisions and misunderstandings — crucial in finance, healthcare, and research.

2 Completeness

Complete information contains all the necessary details required for decision-making; missing data can lead to incomplete analysis and wrong conclusions.

3 Relevance

Information should be useful and applicable to the current problem or decision. Irrelevant data causes confusion and wastes time.

4 Timeliness

Timely information is available when needed. Outdated information can result in missed opportunities, since its value often decreases over time.

Quality of Information (II)

5 Consistency

Consistent information remains the same across multiple sources and does not contradict itself; conflicting reports lower trust in the data.

6 Reliability

Reliable information comes from trustworthy sources and can be depended upon for accurate decision-making — the source plays a crucial role here.

7 Accessibility

Information should be easy for authorized users to obtain. If it is difficult to access or retrieve, it loses its effectiveness.

8 Security

Sensitive or confidential information must be protected from unauthorized access, modification, or destruction, preventing fraud and data breaches.

BY PURPOSE & USAGE

Types of Information (I)

1 Strategic

Used by top management for long-term planning, goal-setting, and decision-making. It focuses on the organization's overall direction and is usually external and forward-looking.

e.g. *A company analyzing market trends for a five-year expansion plan.*

2 Tactical

Used by middle management to develop strategies aligned with overall goals, focusing on medium-term planning, resource allocation, and process optimization.

e.g. *A sales manager adjusting marketing strategy from regional data.*

3 Operational

Used for day-to-day activities and short-term decision-making, helping routine processes run smoothly. This information is often structured and frequently updated.

e.g. *A cashier checking inventory before placing an order.*

4 Statistical

Numerical data collected, analyzed, and presented to describe trends, patterns, and relationships — widely used in forecasting and performance evaluation.

e.g. *Monthly sales figures, unemployment rates, inflation statistics.*

Types of Information (II)

5 Internal

Originates within an organization and is used for decision-making, planning, and performance tracking; usually confidential to the organization.

e.g. *Employee productivity reports; internal financial budgeting records.*

6 External

Comes from sources outside an organization, helping businesses understand market trends, competition, and regulations, and adapt to external change.

e.g. *A company analyzing a competitor's pricing strategy.*

7 Real-Time

Updated instantly and used for immediate decision-making; critical in industries where timely responses are necessary.

e.g. *Live stock market updates; GPS navigation with real-time traffic.*

Source of Information (I)

1 Primary Source

Provides direct, first-hand information that has not been altered or interpreted by anyone else — the most authentic type of source.

e.g. *Original research papers; eyewitness accounts; census data.*

2 Secondary Source

Interprets, analyzes, or summarizes information from primary sources; useful for a broader understanding but may introduce bias.

e.g. *Textbooks summarizing discoveries; review articles.*

3 Tertiary Source

Compiles and organizes information from primary and secondary sources, typically used for quick reference without original research.

e.g. *Encyclopedias; dictionaries; bibliographies.*

4 Internal Source

Originates within an organization or institution and is used for internal decision-making, evaluation, and operational management.

e.g. *Company financial reports; employee performance records.*

Source of Information (II)

5 External Source

Comes from outside an organization and provides independent perspectives, industry trends, and market analysis to stay competitive.

e.g. *News articles; market research reports; government publications.*

6 Formal Source

Follows a structured process ensuring accuracy, credibility, and verification — often peer-reviewed or officially recognized.

e.g. *Academic journals; legal documents; official press releases.*

7 Informal Source

Provides unverified or loosely structured information, useful for gathering opinions and trends but requiring careful validation.

e.g. *Social media posts; blogs; word-of-mouth recommendations.*

COMPARISON

System Analysis vs. System Design

Feature	System Analysis	System Design
Focus	Understanding the problem domain and gathering requirements	Specifying the solution and designing the system
Output	Requirements document	System design specifications
Techniques	Interviews, surveys, observation, document analysis	UML, DFDs, ERDs, decision trees, state transitions
Level of Detail	High-level understanding	Detailed specifications

The output of the analysis phase (the requirements document) becomes the direct input to the design phase — the two are tightly interconnected, and design specifications must align with the requirements to meet stakeholder needs.

CORE CONCEPT

The Information System (IS)

An Information System (IS) is a structured combination of hardware, software, data, people, and processes that work together to collect, store, process, and distribute information. It is designed to support decision-making, coordination, analysis, and control within an organization.

Examples

- Banking system — processes customer transactions such as deposits and transfers
- Inventory management system — tracks product stock levels in real time
- Healthcare system — manages patient records for accurate, timely care

These systems enable organizations to operate efficiently by ensuring accurate data processing and timely information availability.

Why Is an Information System Necessary?

1 Saves Time

An information system helps businesses work faster and stay organized. It saves time by automating routine tasks like tracking sales, managing employees, or storing customer details.

2 Better Decision-Making

It helps in decision-making by providing accurate and updated information. Managers can use it to analyze business performance and plan effectively for the future.

3 Better Communication

Employees can share data easily, work together, and serve customers more efficiently, improving coordination across the organization.

4 Keeps Data Safe

It keeps data safe by preventing unauthorized access and cyber threats — making an organization's information more secure overall.

Types of Information Systems (I)

1 Transaction Processing System (TPS)

Designed to handle and record routine, day-to-day business transactions such as sales, payroll, and order processing. A TPS must be reliable, fast, and accurate.

e.g. *Banking withdrawals; retail POS sales; online booking systems.*

2 Management Information System (MIS)

Collects and processes data from TPS and other sources to generate reports that help managers monitor business performance for planning and decisions.

e.g. *Sales reports; employee performance reports; inventory monitoring.*

3 Decision Support System (DSS)

Helps managers and executives make informed decisions by analyzing large amounts of data and providing insights, forecasts, and recommendations.

e.g. *Stock market analysis; medical diagnosis systems; supply chain optimization.*

4 Enterprise Resource Planning (ERP)

Integrates different business functions like finance, HR, sales, and supply chain into a single system, letting departments share data in real time.

e.g. *SAP ERP; Oracle ERP; a University ERP for admissions and payroll.*

Types of Information Systems (II)

1 Customer Relationship Management (CRM)

Manages customer interactions, tracks sales, and improves service — helping businesses build stronger client relationships.

e.g. Salesforce CRM; HubSpot CRM; Call Center CRM.

2 Supply Chain Management (SCM)

Helps businesses manage the flow of goods, services, and information across the supply chain for efficient delivery.

e.g. Amazon's SCM system; Walmart's inventory restocking; logistics tracking.

3 Expert System (ES)

Mimics human decision-making using artificial intelligence and knowledge-based rules to solve specialized problems.

e.g. Medical expert systems; financial fraud detection; engineering design systems.

Different information systems serve different levels of an organization — from daily transactions to strategic decisions — helping businesses stay efficient and competitive.